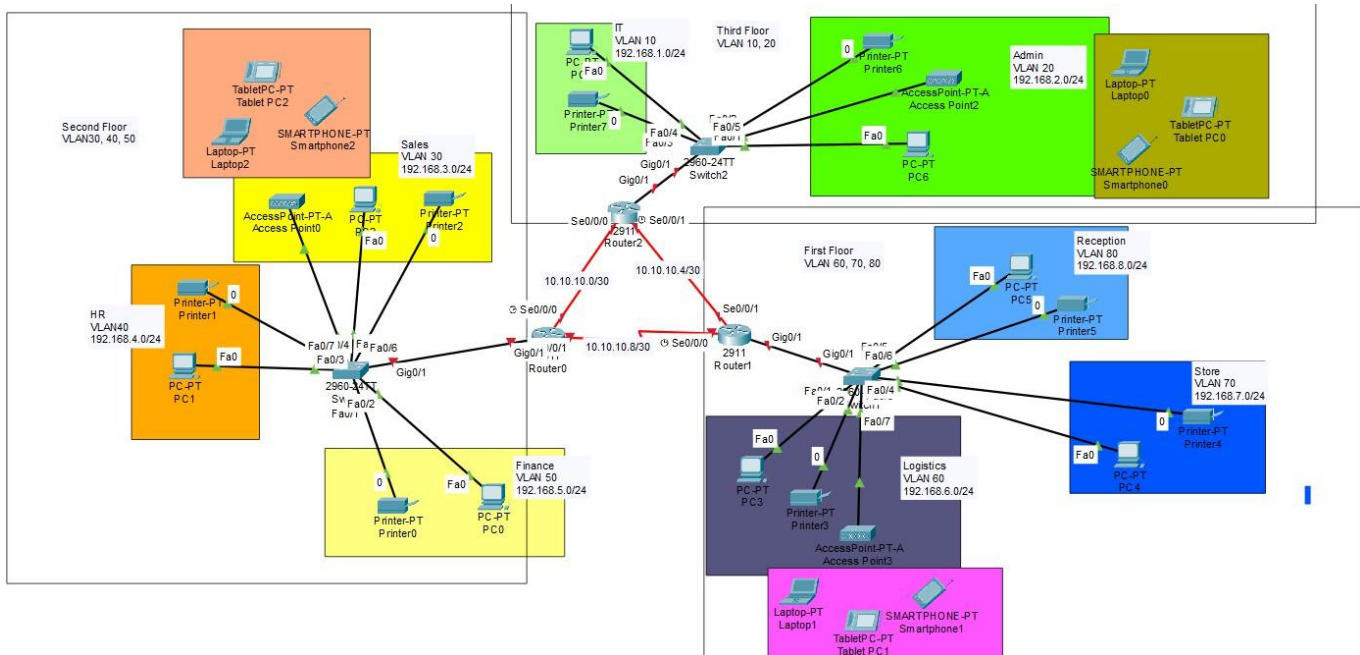




You are required to design and implement Modern Hotel network. The hotel has three floors; in the first floor there three departments (Reception, store and Logistics), in the second floor there are three departments (Finance, HR and Sales/Marketing), while the third floor hosts the IT and Admin. Therefore, the following are part of the considerations during the design and implementation:

- There should be three routers connecting each floor (all placed in the server room in IT department).
- All routers should be connected to each other using serial DCE cable. (Use The WIC named HWIC-2T to connect serial)
- The network between the routers should be 10.10.10.0/30, 10.10.10.4/30 and 10.10.10.8/30.
- Each floor is expected to have one switch (placed in the respective floor).
- Each floor is expected to have WIFI networks connected to laptops and phones.
- Each department is expected to have a printer.
- Each department is expected to be in different VLAN with the following details;

1st Floor;

- Reception- VLAN 80, Network of 192.168.8.0/24
- Store- VLAN 70, Network of 192.168.7.0/24
- Logistics- VLAN 60, Network of 192.168.6.0/24

2nd Floor;

- Finance- VLAN 50, Network of 192.168.5.0/24
- HR- VLAN 40, Network of 192.168.4.0/24

- Sales- VLAN 30, Network of 192.168.3.0/24

3rd Floor;

- Admin- VLAN 20, Network of 192.168.2.0/24

- IT- VLAN 10, Network of 192.168.1.0/24

- Use OSPF as the routing protocol to advertise routes.
- All devices in the network are expected to obtain IP address dynamically with their respective router configured as the DHCP server.
- All the devices in the network are expected to communicate with each other.
- Configure SSH in all the routers for remote login.
- In IT department, add PC called Test-PC to port fa0/20 and use it to test remote login.
- In IT department, add Server called hotel-server and use it as the NTP Server for all routers and switches on the network
- Block any ping from HR and Sales Vlan's to the hotel server
- Allow only https from stores to hotel server
- Configure port security to IT-dept switch to allow only Test-PC to access port fa0/20 (use sticky method to obtain mac-address with violation mode of shutdown.)

You are required to hand in:

1. Packet tracer file containing your project
2. A pdf containing screenshots with the results of these commands:
 - The Topology you have built
 - **Show vlan** command on every switch
 - **Show ip interface brief** command on all switches and routers
 - **Show running-config** on all routers
 - A screenshot from test-pc command prompt making ssh to all routers and switches
 - **show access-lists** from routers where you applied access list
 - **A screenshot from a device from every vlan with its dhcp ip**